

Evolutionary certificate search

grammar GP, SOS/LP, and an adversarial grid on ψ_- , $\lambda_2(C) - 1$, $q^\dagger$, and $\text{ind}_-(A_0)$

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Abstract

Round 438 is a sealed *search* round, not a lemma-first grind: residual-0 finds become theorems, but nothing counts without a machine-precision gate on every training window, a hard train/holdout split, $\mathbb{Q}$ -exact toys, and kill tests. The r395 truncation artefact is why holdout worlds (χ_3/χ_4 , EXT, Selected $k \geq 7$) are never seen in any fitness loop. Three independent tracks share the r407/r431/r433 builders. Ausgang **A:NOT_FOUND** / **B:NOT_FOUND** / **C:NOT_FOUND**: no exact source formula for ψ_- , $\lambda_2(C) - 1$ or $q^\dagger$; no source-pure SOS certificate for $-\det K_2$ or $\text{ind}_-(A_0) \leq 1$; no canonical-cut window with $\text{ind}_-(A_0) \geq 2$. Truncation $n = N_w - 5$ *does* inflate $\#\{\lambda(C) < 1\}$ to 2 — typed r395, not a falsifier of the two-case statement at the canonical cut $n = N_w - 3$. No RH claim.

Status. Lemmas 1–2 are proved as finite identities. Propositions 1–3 are sealed search outcomes (honest negatives). No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The split (r395)

Fitness of every track sees only MAIN windows $kz \in \{9, 18, 20\}$ and, for Track C, the sealed canonical-adjacent MAIN grid $\{5, 9, 12, 16, 18, 20, 22, 36, 38, 39, 52\}$ at the working depth $n = N_w - 3$. The following are **never** in a fitness loop: χ_3 and χ_4 (matched frame, parity $a = 1$), the EXT anchor $kz = 35$, and Selected $k \geq 7$ ($kz = 43$). They are evaluated after search. Fractional arch-parity $a \notin \{0, 1\}$ destroys the Gamma factor (r357) and is a control, not a candidate.

Sealed budget: Track A seed 43801, population 24, 10 generations, depth 4 (smoke: 8×2); Track B seed 43802, 250 projected-gradient steps; Track C seed 43803, enumerated 11×5 depth grid plus a 72-point weight-scale GA on MAIN-9 only. Smoke < 60 s (measured 0.4 s); full $\ll 30$ min (measured 2.0 s).

2 $\mathbb{Q}$ toys

Lemma 1 (machinery). *On the two-node toy $Y = \{0, 1\}$, the grammar expression $1 - Y$ recovers e_0 with cosine 1 (residual 0). The r375 factorization $\det K_2 = -7$, $\gamma = 9/2$, $\lambda_- = 1$ is exact. $A_0 = \text{diag}(-1, 1, 2, 3) = \text{diag}(1, 1, 2, 3) - 2e_0e_0^T$ has remainder 0 and $\text{ind}_-(A_0) = 1$. The detector flags $\text{diag}(-1, -1/2, 2, 3)$ as $\text{ind}_- = 2$.*

Lemma 2 (train/holdout ward). *The fitness tag set is exactly $\{\text{MAIN-9}, \text{MAIN-18}, \text{MAIN-20}\}$. Permute of MAIN-9 has $n_C = 21$; scramble has $n_C = 21$. The illegal hybrid χ_3 with $a = 0$ has $n_C = 3$.*

3 Track A: grammar GP

Hole samples of ψ_- are the bottom eigenvector of the r407 matrix C (already on Y). Grammar terminals: $y_i, u^\vee(y_i), P'_Y(y_i), 1/P'_Y, \sqrt{u^\vee}, \mathbf{1}$, Nyquist face, $\theta_i = \arccos y_i$, $\text{sign}(P'_Y)$, hole gap, distance to nearest X -node.

Proposition 1 (NOT_FOUND). *On MAIN-9 the four canonical candidates have $|\cos| =$ ones 0.0072, centred 0.0010, Nyquist 0.3859, v_{top} 0.6766. Uniformly on $\{9, 18, 20\}$ the best seed is $1/P'_Y$ (min-cos 0.1010); v_{top} drops to 0.0584 at $\text{kz} = 18$. Sealed GP (seed 43801, 24×10) stays at min-cos 0.1010, expression INVPY. Source NNLS for $\lambda_2(C) - 1$ and $q^\dagger$ has relative residual $\gg 1$ (predictions not even signed correctly). Holdout min-cos of INVPY on $n_C = 1$ worlds is not 0.999.*

The target > 0.999 uniform is missed by an order of magnitude. No FOUND_CANDIDATE (that bar is 0.9).

4 Track B: SOS/LP

Proposition 2 (NOT_FOUND). *On the $n_- = 1$ train branch, nonnegative least squares of eight source-pure scalars onto $-\det K_2$ has relative residual 13.2 (coefficients cannot represent the target). The r375 identity $-\det K_2 = \det(I + R_+)(\gamma/\lambda_- - 1)$ holds on MAIN-9 (fact $= -5.038869 = \det K_2$, $\gamma/\lambda_- = 1.846$) but consumes ψ_- and is therefore not a source certificate. The family $P = \theta_0 I + \theta_1 \text{diag}(u^\vee)$, $v = v_{\text{top}}$, remainder $A_0 - P + vv^T$ has best $\lambda_{\min} = -0.261$ (not PSD) on the sealed 6×6 grid.*

5 Track C: adversarial grid

The two-case object is $A_0 = R_{N_w-3} - I/2$, equivalently $n_C = \#\{\lambda(C) < 1\}$ at depth $n = N_w - 3$.

Proposition 3 (NOT_FOUND, truncation diagnostic). *On the canonical cut $dn = 0$, all 11 CORE windows have $n_C \leq 1$. Closest $\lambda_2(C) - 1 = 2.245 \cdot 10^{-7}$ at $\text{kz} = 52$. Weight-scale GA ($|\varepsilon| \leq 2 \cdot 10^{-4}$, 72 points on MAIN-9) keeps $n_C = 1$. Holdout: χ_3 -9 $n_C = 0$, χ_3 -15 $n_C = 1$, χ_3 -19 $n_C = 1$, χ_4 -9 $n_C = 1$, χ_4 -20 $n_C = 0$, Selected $k = 7$ $n_C = 0$, EXT-35 $n_C = 1$ — none has $n_C \geq 2$. Truncation $dn = -2$ inflates $n_C = 2$ on $\text{kz} \in \{5, 18, 22, 38, 52\}$ (and $dn = -1$ at $\text{kz} = 52$). That object is not A_0 ; it is the r395 truncation class and is not a falsifier of the two-case statement.*

6 Kill table

world	n_C	$\lambda_2(C)$	role
MAIN-9 (train)	1	1.00017632	fitness
MAIN-18 (train)	1	1.00003201	fitness
MAIN-20 (train)	1	1.00003567	fitness
χ_3 -9 (holdout)	0	1.00542	holdout
Selected $k = 7$ (holdout)	0	1.00000148	holdout
EXT-35 (holdout)	1	1.00000033	holdout
Λ -permute	21	0.0031	kill
scramble	21	0.0051	kill
χ_3 $a = 0$ (illegal)	3	—	geometry kill
MAIN-18, $n = N_w - 5$	2	0.860	r395 truncation

7 What this does not say

The round does not prove $\text{ind}_-(A_0) \leq 1$, does not produce a source formula for ψ_- or $q^\dagger$, and does not falsify the two-case candidate on the canonical cut. It hardens the census: the closest canonical $\lambda_2 - 1$ in the sealed grid is $2.245 \cdot 10^{-7}$, and every holdout world stays in $\{0, 1\}$ negatives. Search was allowed to be dirty; verification was not.

Companion probe `evolutionary_certificate_probe.py` (29/29 full, 29/29 smoke). Verifier `verify_evolutionary_cert.py`. No RH claim.